Q.No :9

DS	Session	Linked List	Question Information	Level 1 Challenge 39

Question description

sanam's Dream came true after he got an Appointment order from Google.Simon's family was very happy of his achievement.

The company mentioned Basic Salary, DA, HRA with some other benefits.

But not highlighted the Gross salary in the order.

sanam's father wanted to know the Gross salary of his son.

sanam try to his gross salary from HR department. they informed that you have to get pass grade in first month entry test. the entry test has 5 questions. one of the question was, Split a circular linked list in two halves. you have to split the circular Linked List with the same size of Divisions.Maybe if circular Linked List is odd, you have to change the number of node, it is even .

Can you help sanam?

Function Description

First count the number of node in Circular Linked List.

Second, you have to make the List even.

Third, you need to make the List half. the front is the same size like the rear. Finally, I have to make two circular Linked List.

Constraints

0<n<10

Input Format:
The First line represents the number of input in the circular linked list elements

Output Format:
First Line indicates the complete linked list

second line indicates the odd list

third line indicates the even list

Refer sample test cases.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
6
```

EXPECTED OUTPUT

```
Complete linked_list:
[h]=>1=>2=>3=>4=>5=>6=>[h]
Odd:
[h]=>1=>3=>5=>[h]
Even:
[h]=>2=>4=>6=>[h]
```

Test Case 2

INPUT (STDIN)

```
15
```

EXPECTED OUTPUT

```
Complete linked_list:
[h]=>1=>2=>3=>4=>5=>6=>7=>8=>9=>10=>11=>12=>13=>14=>15=
>[h]
Odd:
[h]=>1=>3=>5=>7=>9=>11=>13=>15=>[h]
Even:
[h]=>2=>4=>6=>8=>10=>12=>14=>[h]
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
void insert(int data)
```

Test Case 2

KEYWORD

```
struct n *next;
```

Test Case 3

KEYWORD

```
void display(struct n *h)
```

Complexity Test Cases

Code:

```
#include <stdio.h>
#include <stdlib.h>

struct n
{
int data;
struct n *next;
};

struct n *head = NULL;

void insert(int data)
{
struct n *newnode;
```

```
struct n *temp;

newnode = (struct n *)malloc(sizeof(struct n));

newnode->data = data;
newnode->next = NULL;

if(head == NULL)
{
head = newnode;
newnode->next = head;
}
else
{
temp = head;

while(temp->next != head)
{
temp = temp->next;
}

temp->next = newnode;
newnode->next = head;
}
}

void display(struct n *h)
{
struct n *temp;

if(h == NULL)
return;

temp = h;

do
{
printf("%d", temp->data);
temp = temp->next;

if(temp != h)
printf("=>");

} while(temp != h);
}

void displayOdd(struct n *h, int count)
{
int i;
struct n *temp = h;

printf("Odd:\n");
printf("[h]=>");
```

```c
for(i = 1; i <= count; i++)
{
if(i % 2 != 0)
{
printf("%d", temp->data);

if(i < count - 1)
printf("=>");
}

temp = temp->next;
}

printf("=>[h]\n");
}

void displayEven(struct n *h, int count)
{
int i;
struct n *temp = h;

printf("Even:\n");
printf("[h]=>");

for(i = 1; i <= count; i++)
{
if(i % 2 == 0)
{
printf("%d", temp->data);

if(i < count)
printf("=>");
}

temp = temp->next;
}

printf("=>[h]\n");
}

int main()
{
int n;
int i;

printf("Input value:\n");
scanf("%d", &n);

for(i = 1; i <= n; i++)
{
insert(i);
}
```

```
printf("\nOutput:\n");

printf("Complete linked_list:\n");
printf("[h]=>");
display(head);
printf("=>[h]\n");

displayOdd(head, n);
displayEven(head, n);

return 0;
}
```

Output:

Test_case 1:

```
Input value:
6
Complete linked_list:
[h]=>1=>2=>3=>4=>5=>6=>[h]
Odd:
[h]=>1=>3=>5=>[h]
Even:
[h]=>2=>4=>6=>[h]
```

Test_case 2:

```
Input value:
15
Complete linked_list:
[h]=>1=>2=>3=>4=>5=>6=>7=>8=>9=>10=>11=>12=>13=>14=>15=>[h]
Odd:
[h]=>1=>3=>5=>7=>9=>11=>13=>15=>[h]
Even:
[h]=>2=>4=>6=>8=>10=>12=>14=>[h]
```